

Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.34_e-1)

Metric

$$\begin{aligned} ds^2 = & dz \otimes dz + \left(-\frac{576}{R^2 w^4 + R^2 x^4 + R^2 y^4 - 48 R w^2 - 2 (R^2 w^2 - 24 R) x^2 - 2 (R^2 w^2 - R^2 x^2 - 24 R) y^2 + 576} \right) dw \\ & \otimes dw + \left(\frac{576}{R^2 w^4 + R^2 x^4 + R^2 y^4 - 48 R w^2 - 2 (R^2 w^2 - 24 R) x^2 - 2 (R^2 w^2 - R^2 x^2 - 24 R) y^2 + 576} \right) dx \\ & \otimes dx + \left(\frac{576}{R^2 w^4 + R^2 x^4 + R^2 y^4 - 48 R w^2 - 2 (R^2 w^2 - 24 R) x^2 - 2 (R^2 w^2 - R^2 x^2 - 24 R) y^2 + 576} \right) dy \otimes dy \end{aligned}$$

Coordinates

$$(x^\mu) = (z, w, x, y)$$

Null Tetrad

Covariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dz + \left(-\frac{12 \sqrt{2}}{R w^2 - R x^2 - R y^2 - 24} \right) dw$$

$$n = \left(\frac{1}{2} \sqrt{2} \right) dz + \left(-\frac{12 \sqrt{2}}{R w^2 - R x^2 - R y^2 - 24} \right) dw$$

$$m = \left(-\frac{12 \sqrt{2}}{R w^2 - R x^2 - R y^2 - 24} \right) dx + \left(-\frac{12i \sqrt{2}}{R w^2 - R x^2 - R y^2 - 24} \right) dy$$

$$\bar{m} = \left(-\frac{12 \sqrt{2}}{R w^2 - R x^2 - R y^2 - 24} \right) dx + \left(\frac{12i \sqrt{2}}{R w^2 - R x^2 - R y^2 - 24} \right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dz + \left(\frac{1}{48}\sqrt{2}Rw^2 - \frac{1}{48}\sqrt{2}Rx^2 - \frac{1}{48}\sqrt{2}Ry^2 - \frac{1}{2}\sqrt{2}\right) dw$$

$$n = \left(\frac{1}{2}\sqrt{2}\right) dz + \left(\frac{1}{48}\sqrt{2}Rw^2 - \frac{1}{48}\sqrt{2}Rx^2 - \frac{1}{48}\sqrt{2}Ry^2 - \frac{1}{2}\sqrt{2}\right) dw$$

$$m = \left(-\frac{1}{48}\sqrt{2}Rw^2 + \frac{1}{48}\sqrt{2}Rx^2 + \frac{1}{48}\sqrt{2}Ry^2 + \frac{1}{2}\sqrt{2}\right) dx + \left(-\frac{1}{48}i\sqrt{2}Rw^2 + \frac{1}{48}i\sqrt{2}Rx^2 + \frac{1}{48}i\sqrt{2}Ry^2 + \frac{1}{2}i\sqrt{2}\right) dy$$

$$\bar{m} = \left(-\frac{1}{48}\sqrt{2}Rw^2 + \frac{1}{48}\sqrt{2}Rx^2 + \frac{1}{48}\sqrt{2}Ry^2 + \frac{1}{2}\sqrt{2}\right) dx + \left(\frac{1}{48}i\sqrt{2}Rw^2 - \frac{1}{48}i\sqrt{2}Rx^2 - \frac{1}{48}i\sqrt{2}Ry^2 - \frac{1}{2}i\sqrt{2}\right) dy$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\partial_z + \frac{1}{48}\sqrt{2}Rw^2 - \frac{1}{48}\sqrt{2}Rx^2 - \frac{1}{48}\sqrt{2}Ry^2 - \frac{1}{2}\sqrt{2}\partial_w$$

$$\Delta = \frac{1}{2}\sqrt{2}\partial_z + \frac{1}{48}\sqrt{2}Rw^2 - \frac{1}{48}\sqrt{2}Rx^2 - \frac{1}{48}\sqrt{2}Ry^2 - \frac{1}{2}\sqrt{2}\partial_w$$

$$\delta = -\frac{1}{48}\sqrt{2}Rw^2 + \frac{1}{48}\sqrt{2}Rx^2 + \frac{1}{48}\sqrt{2}Ry^2 + \frac{1}{2}\sqrt{2}\partial_x - \frac{1}{48}i\sqrt{2}Rw^2 + \frac{1}{48}i\sqrt{2}Rx^2 + \frac{1}{48}i\sqrt{2}Ry^2 + \frac{1}{2}i\sqrt{2}\partial_y$$

$$\bar{\delta} = -\frac{1}{48}\sqrt{2}Rw^2 + \frac{1}{48}\sqrt{2}Rx^2 + \frac{1}{48}\sqrt{2}Ry^2 + \frac{1}{2}\sqrt{2}\partial_x + \frac{1}{48}i\sqrt{2}Rw^2 - \frac{1}{48}i\sqrt{2}Rx^2 - \frac{1}{48}i\sqrt{2}Ry^2 - \frac{1}{2}i\sqrt{2}\partial_y$$

Spin Coefficients

$$\rho = \frac{1}{24}\sqrt{2}Rw$$

$$\sigma = 0$$

$$\kappa = \frac{1}{48} \sqrt{2} R x + \frac{1}{48} i \sqrt{2} R y$$

$$\mu = -\frac{1}{24} \sqrt{2} R w$$

$$\lambda = 0$$

$$\nu = -\frac{1}{48} \sqrt{2} R x + \frac{1}{48} i \sqrt{2} R y$$

$$\alpha = \frac{1}{48} \sqrt{2} R x - \frac{1}{48} i \sqrt{2} R y$$

$$\beta = -\frac{1}{48} \sqrt{2} R x - \frac{1}{48} i \sqrt{2} R y$$

$$\tau = \frac{1}{48} \sqrt{2} R x + \frac{1}{48} i \sqrt{2} R y$$

$$\pi = -\frac{1}{48} \sqrt{2} R x + \frac{1}{48} i \sqrt{2} R y$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{1}{12} R$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{24} R$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{1}{12} R$$

$$\Lambda = \frac{1}{24} R$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"